SIMATS ENGINEERING

ITA 06 – MACHINE LEARNING | Assessment Tool 2 (CO1) – Annexure A

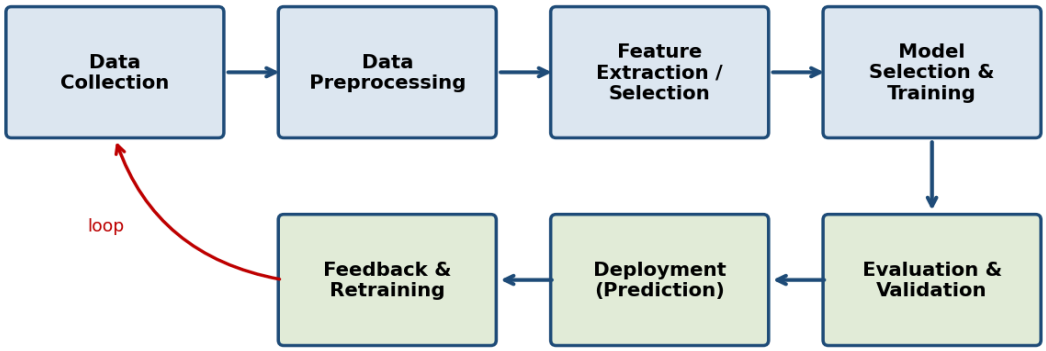
Concept Mapping – Answers to Questions 1 to 5

Q1. Blocks of a Machine Learning System; Supervised and Unsupervised Examples (5 Marks)

1. Concept

A machine learning system is a chain of blocks that converts raw data into a trained model which makes predictions on new data.

2. Block diagram



3. Explanation of the blocks

- **Data collection:** gathering raw data from sensors, databases, web, surveys, etc.
- **Data preprocessing:** cleaning noise, handling missing values, removing duplicates, normalising/encoding.
- **Feature extraction / selection:** choosing the most useful attributes and reducing dimensionality.
- **Model selection and training:** choosing an algorithm and learning its parameters from the training data.
- **Evaluation and validation:** testing on unseen data using metrics (accuracy, precision, recall, loss); tuning hyper-parameters.
- **Deployment (prediction):** using the final model on new, real-world inputs.
- **Feedback and retraining:** monitoring performance and updating the model with new data.

4. Supervised vs Unsupervised learning

Point	Supervised Learning	Unsupervised Learning
Data	Labelled (input + known output)	Unlabelled (input only)
Goal	Learn mapping from input to output	Discover hidden structure/patterns
Types	Classification, Regression	Clustering, Association, Dimensionality reduction
Example	Spam filter: emails labelled Spam / Not Spam train a classifier; house price prediction	Customer segmentation: K-Means groups customers by buying behaviour with no labels

Conclusion: All the above blocks work together in a loop; supervised learning uses labelled data to predict, while unsupervised learning finds groups in unlabelled data.

Q2. ML System for Disaster Management (Floods and Earthquakes) (5 Marks)

1. Concept

The system uses historical records (past floods, earthquakes) and real-time data to predict the event, its location and severity, so authorities can act early.

2. Design

Component	Choice / Technique
Data sources	Historical: rainfall, river level, seismic records, satellite images. Real-time: rain gauges, river-level sensors, seismometers, IoT, weather feeds
Data preprocessing	Remove noise and outliers, fill missing values (interpolation), normalise units/scales, align time stamps, balance rare-event data (SMOTE)
Feature selection	Correlation analysis, Random Forest feature importance, PCA/RFE. Floods: rainfall, soil moisture, river level, elevation. Earthquakes: magnitude, depth, past tremor frequency, fault-line distance
Choice of model	Random Forest / Gradient Boosting for tabular data (robust, handles non-linearity); LSTM for time-series (river level, seismic signals); CNN for satellite images

(a) Training, validation and evaluation steps

- Split data into training (70%), validation (15%) and test (15%) sets (time-based split for time-series).
- Train the model on the training set.
- Validate: tune hyper-parameters using the validation set and k-fold cross-validation; use early stopping to avoid overfitting.
- Evaluate on the test set using recall (most important – missing a disaster is costly), precision, F1-score, ROC-AUC and RMSE for severity.
- Deploy, monitor and retrain with new data.

(b) Early warning and emergency response

- Real-time predictions trigger alerts (SMS, sirens, mobile apps) to people in risk zones before the event.
- Risk maps guide evacuation routes and the placement of shelters, rescue teams and relief supplies.
- Authorities can shut down dams, trains and power in advance, reducing loss of life and property.

Conclusion: A high-recall ensemble/LSTM model with good preprocessing and feature selection gives reliable early warning for disaster management.

Q3. Find-S Algorithm – Play Tennis (5 Marks)

1. Concept

Find-S starts with the most specific hypothesis h = <∅,∅,∅,∅>, considers only positive examples, and generalises an attribute to '?' whenever the example value differs. Negative examples are ignored.

2. Positive examples (Play Tennis = Yes)

Examples 3, 4, 5 and 7. Examples 1, 2, 6 and 8 are negative and are ignored.

3. Step-by-step updates

Step	Example	Hypothesis h (Outlook, Temp, Humidity, Wind)
Initial	–	<∅, ∅, ∅, ∅>
1	Ex 3: Overcast, Hot, High, Weak	<Overcast, Hot, High, Weak>
2	Ex 4: Rainy, Mild, High, Weak	<?, ?, High, Weak>
3	Ex 5: Rainy, Cool, Normal, Weak	<?, ?, ?, Weak>
4	Ex 7: Overcast, Mild, Normal, Strong	<?, ?, ?, ?>

4. Final Answer

Maximally specific hypothesis: h = <?, ?, ?, ?>

Interpretation: the positive examples differ in every attribute, so Find-S generalises completely and accepts every day. This shows the weakness of Find-S – it cannot represent this concept with a conjunctive hypothesis and it ignores negative examples.

Q4. Candidate Elimination Algorithm – Enjoy Sport (5 Marks)

1. Concept

Candidate Elimination keeps both boundaries of the version space. S (specific) is generalised by positive examples; G (general) is specialised by negative examples. Inconsistent hypotheses are removed.

Initial: S0 = {<∅,∅,∅,∅>}, G0 = {<?,?,?,?>}

2. Step-by-step updates (Weather, Temperature, Humidity, Wind)

Ex	Training example	S boundary	G boundary
1	Sunny, Warm, Normal, Weak (+)	S1 = {<Sunny, Warm, Normal, Weak>}	G1 = {<?, ?, ?, ?>}
2	Sunny, Warm, High, Strong (+)	S2 = {<Sunny, Warm, ?, ?>}	G2 = {<?, ?, ?, ?>}
3	Rainy, Cold, High, Weak (–)	S3 = {<Sunny, Warm, ?, ?>}	G3 = {<Sunny, ?, ?, ?>, <?, Warm, ?, ?>}
4	Sunny, Cold, Normal, Weak (+)	S4 = {<Sunny, ?, ?, ?>}	G4 = {<Sunny, ?, ?, ?>} (<?, Warm, ?, ?> removed – it does not cover Ex 4)
5	Rainy, Warm, Normal, Strong (–)	S5 = {<Sunny, ?, ?, ?>}	G5 = {<Sunny, ?, ?, ?>} (already excludes Rainy)

3. Final Answer

S = {<Sunny, ?, ?, ?>} and G = {<Sunny, ?, ?, ?>}

Since S = G, the version space has converged to a single hypothesis: the person enjoys the sport when Weather = Sunny.

Q5. Find-S Algorithm – Customer Buys Product (5 Marks)

1. Concept

Find-S begins with the most specific hypothesis and generalises it using only the positive examples (Buys Product = Yes).

2. Positive examples

Examples 3 and 4. Examples 1 and 2 are negative and are ignored.

3. Step-by-step updates

Step	Example	Hypothesis h (Age, Income, Student, Credit Rating)
Initial	–	<∅, ∅, ∅, ∅>
1	Ex 3: Middle, High, No, Fair	<Middle, High, No, Fair>
2	Ex 4: Senior, Medium, No, Fair	<?, ?, No, Fair>

4. Final Answer

Maximally specific hypothesis: h = <?, ?, No, Fair>

Interpretation: a customer buys the product when he/she is not a student and has a Fair credit rating, whatever the age and income.

Rubric Coverage Summary (Rubrics for Calculation)

Rubric Criteria	Weightage	How it is addressed in the answers
Conceptual Understanding	30	Concept stated first in every answer (ML blocks, Find-S, Candidate Elimination)
Accuracy of Answer	20	Correct final results: <?,?,?,?>, S = G = <Sunny,?,?,?>, <?,?,No,Fair>
Application of Knowledge	20	Algorithms applied step by step to the given datasets
Completeness of Response	15	All parts (a, b, diagram, examples) of every question answered
Clarity & Presentation	15	Diagram, tables, steps and bold final answers